

Inference on Union Bounds with Applications to DiD, RDD, Bunching, and Structural Counterfactuals

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September 29, 2023

Abstract

A union bound is a union of multiple bounds. Union bounds occur in a wide variety of empirical settings, from relaxations of the difference-in-differences parallel trends assumption to counterfactual analysis with partially identified structural parameters. In this paper I provide the first general and systematic study of inference on these kinds of bounds. When the union is taken over a finite set, I propose a confidence interval based on modified conditional inference. I show that it improves upon existing methods in a large set of data generating processes. When the union is taken over an infinite set, I assume the set is defined by moment inequalities, as is common in practice. I then propose a calibrated projection based inference procedure that generalizes results from the moment inequality subvector inference literature and is computationally simple. Finally, the new procedures give statistically significant results while the pre-existing alternatives do not in two empirical applications, the sensitivity analysis in [Dustmann, Lindner, Schönberg, Umkehrer, and Vom Berge \(2022\)](#) and the counterfactual analysis in [Dickstein and Morales \(2018\)](#).

KEYWORDS: Bound analysis, partial identification, conditional inference, moment inequalities, sensitivity analysis, counterfactual analysis

JEL classification: C01, C12, C15